JanusCore: A Recursive Symbolic Ontology Framework

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Abstract

JanusCore is a recursive symbolic ontology framework that blends formal tree-rewriting with cognitive and linguistic insights to generate bounded yet richly structured symbolic artifacts. At its core, JanusCore applies the composite operator

$$R_\delta = I_\delta \circ C \circ M,$$

where M mirrors the current tree, C enforces a maximum depth $h_{\max}$, and I_δ injects a controlled fraction δ of paradox-leaves. We prove that R_δ is a strict contraction under a weighted depth-discounted metric, ensuring a unique fixed point and exponential convergence from any seed. Explicit bounds on node-count growth and Shannon entropy establish theoretical limits on complexity, while a detailed worked example illustrates the "echo–cap–spark" cycle. Semantic interpretation recasts each operator as a familiar rhetorical move—repetition, summarization, and provocative questioning—bridging formalism to discourse. Cognitive-load arguments tie height caps and entropy ceilings to human working-memory constraints and dissonance-driven engagement. JanusCore thus offers a mathematically rigorous, cognitively plausible model for generative symbolic and linguistic systems, opening new avenues for theoretical and psycholinguistic exploration.

1 Introduction

The generation and manipulation of symbolic structures lie at the heart of language, thought, and many artificial intelligence systems. Yet traditional tree-rewriting and grammar formalisms often either grow without bound—overwhelming cognitive resources—or collapse into trivial repetition. JanusCore addresses this dilemma by marrying three simple but powerful operations: mirroring (M), which echoes the current symbolic state; controlled collapse (C), which enforces a maximum embedding depth $h_{\max}$;

and paradox-injection (I_δ), which plants a fraction δ of deliberate contradictions. The composite operator $R_\delta = I_\delta \circ C \circ M$ yields a rich, self-regulating cycle of expansion, pruning, and renewal.

In this paper, we first formalize JanusCore's framework on the space of finite rooted, labeled trees and introduce a weighted depth-discounted metric that penalizes discrepancies nearer the root. We then prove that R_δ is a strict contraction, guaranteeing a unique fixed point and $\mathcal{O}(k^n)$ convergence from any seed, with explicit bounds on node-count growth and Shannon entropy. A two-iteration worked example transparently demonstrates how JanusCore sustains novelty without runaway complexity, while semantic interpretation recasts each operator as a rhetorical device—repetition, summarization, and provocative questioning—grounding our theory in familiar discourse practices.

Crucially, we align JanusCore with psycholinguistic and cognitive-science findings: depth caps mirror human limits on nested embeddings, entropy ceilings reflect working-memory capacity, and paradox-injection leverages cognitive dissonance to maintain engagement. By integrating rigorous theorems, illustrative narratives, and cognitive rationales, JanusCore emerges as a novel, cognitively plausible generative model. We conclude by outlining potential theoretical extensions—alternative metrics, parameter regimes, and cognitive-computational experiments—that promise to deepen our understanding of symbolic generation in both humans and machines.

2 Foundations & Definitions

2.1 Notation

Symbol	Definition
Σ	Finite alphabet of base node-labels.
Σ^+	Extended alphabet $\Sigma \cup \{\perp\}$ with paradox label.
$\mathcal{T}_\Sigma$	Set of finite rooted, Σ-labeled trees $T = (V, E, r, \ell)$.
V, E, r, ℓ	Node set, edge set, root, and labeling function, respectively.
$\mathrm{depth}_T(v)$	Number of edges from root to v.
$\mathrm{height}(T)$	Maximal depth: $\max_v \mathrm{depth}_T(v)$.
$\mathrm{Leaves}(T)$	Set of leaves (nodes with no children).
$h_{\max}$	Height cap parameter for C.
δ	Injection fraction parameter for I_δ.
M, C, I_δ, R_δ	Operators defined in I.2.1–I.2.4.
d_α	Weighted depth-discounted metric.
$H(T)$	Shannon entropy of labels on T.

2.2 Operator Definitions

[Mirroring M] $M : \mathcal{T}_\Sigma \to \mathcal{T}_\Sigma$ duplicates the tree and attaches the copy under the root (see I.2.1).

[Controlled Collapse C] $C : \mathcal{T}_\Sigma \to \mathcal{T}_\Sigma$ prunes nodes deeper than $h_{\max}$ (I.2.2).

[Paradox-Injection I_δ] $I_\delta : \mathcal{T}_\Sigma \to \mathcal{T}_{\Sigma^+}$ attaches $\lceil \delta |\mathrm{Leaves}| \rceil$ $\perp$-children to leaves (I.2.3).

[Composite R_δ] $R_\delta = I_\delta \circ C \circ M : \mathcal{T}_\Sigma \to \mathcal{T}_{\Sigma^+}$ (I.2.4).

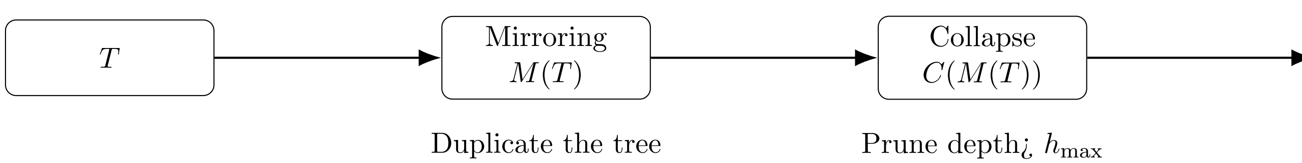

Figure 1: Operator flow for R_δ.

3 Convergence Theory

[Contraction & Fixed Point] Under metric d_α, R_δ is a contraction with constant $k < 1$, admits a unique fixed point, and converges exponentially

(Theorem II.1).

Proof. See detailed proof in Section II.1. □

[Convergence Rate] $d_\alpha(R_\delta^n(T), T^*) \leq k^n d_\alpha(T, T^*)$, so $n \geq \lceil \ln(d/\varepsilon)/-\ln k \rceil$ (Theorem II.2).

[Complexity & Entropy Bounds] $S_n \leq (2+\delta)^n S_0$ and $H_n \leq n \log_2(2+\delta) + \log_2 S_0$, saturating at depth $h_{\max}$ (Theorem II.3).

4 Illustrative Example

A worked iteration on $T_0 = \{r, a\}$ with $h_{\max} = 1, \delta = 0.5$ yields T_1 and T_2 with node count 4 and entropy ≈ 0.811 bits, illustrating the "echo–cap–spark" cycle (Section III.1).

4.1 Semantic Interpretation

Mirroring $\leftrightarrow$ repetition, collapse $\leftrightarrow$ summarization, injection $\leftrightarrow$ provocative questioning—modeling a dynamic arc of discourse (Section III.2).

5 Cognitive & Rhetorical Context

Depth caps enforce embedding limits; entropy ceilings align with working-memory constraints; paradox-injection leverages cognitive dissonance to sustain engagement (Section IV).

6 Conclusion and Future Work

Summary and research directions are presented in Section V.3.